NORDHAVEN

Urban Infrastructure Mapping & Geospatial Data Engineering

Project Type: Independent Geospatial Portfolio Case Study
Study Area: Bremen, Germany
Primary GIS Platform: QGIS 3.10.13
Coordinate Reference System: ETRS89 / UTM Zone 32N (EPSG:25832)

Project Overview

NORDHAVEN is an independently developed geospatial infrastructure mapping case study demonstrating practical capabilities in spatial data preparation, infrastructure digitisation, attribute management, quality validation and cartographic production.

The project uses OpenStreetMap-derived reference data and independently developed demonstration infrastructure layers to establish a structured GIS database and produce a consolidated infrastructure overview map.

Project Objectives

- Develop a structured geospatial database for infrastructure features.
- Organise spatial data using appropriate geometry types and attribute schemas.
- Demonstrate polygon, point and line feature digitisation.
- Apply attribute management and basic spatial quality checks.
- Produce a professional cartographic output using QGIS.

Technical Scope

Workstream	Implementation
Reference data	OpenStreetMap-derived vector layers
GIS database	GeoPackage
Spatial mapping	Polygon, point and line features
Attribute management	Project-specific classification schema
QA/QC	Geometry validation and attribute completeness
Cartography	QGIS print layout and PDF export

Key Skills Demonstrated

QGIS · Geospatial Data Management · Infrastructure Mapping · Spatial Digitisation · Attribute Engineering · GeoPackage · GIS QA/QC · Cartographic Design

Project Status

Completed: Core demonstration database, GIS layers, basic validation and infrastructure overview map.

Future enhancements: Python automation, GeoPandas processing, PostGIS integration and advanced spatial validation.

Independent portfolio case study. Demonstration datasets and outputs are not client deliverables or representations of confidential employer data.